

Engineering Mathematics III (ENGE702)

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Lecture 21: Introduction to Partial Differential Equations (PDEs)

Partial Derivatives

Partial derivatives are used to find rates of change in functions of more than one variable. They describe the rate at which the function is changing *in the direction of the variable*.

For example if we have a function $u(x, y)$, then:

- ▶ $\frac{\partial u}{\partial x}$ is the partial derivative in the x direction,
- ▶ $\frac{\partial u}{\partial y}$ is the partial derivative in the y direction,
- ▶ $\frac{\partial^2 u}{\partial x^2}$ is the partial second derivative in the x direction,
- ▶ $\frac{\partial^2 u}{\partial y^2}$ is the partial second derivative in the y direction,
- ▶ $\frac{\partial^2 u}{\partial x \partial y}$ is the partial derivative in the x direction of $\frac{\partial u}{\partial y}$

Partial Derivatives

Practically, calculating a partial derivative is similar to an ordinary derivative, treat any variable that you are not differentiating like a constant.

Example: If $u(x, y, t) = x \sin(2yt) + 3x^2t$, then calculate:

► $\frac{\partial u}{\partial x},$

► $\frac{\partial u}{\partial t},$

► $\frac{\partial^2 u}{\partial x \partial y}.$

Notation: sometimes we use the simpler notation $u_x = \frac{\partial u}{\partial x}, u_{xy} = \frac{\partial^2 u}{\partial x \partial y},$

$u_{xx} = \frac{\partial^2 u}{\partial x^2}$ etc.

Partial Derivatives

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Example: If $u(x, y, t) = x \sin(2yt) + 3x^2t$, then calculate:

$$\blacktriangleright \frac{\partial u}{\partial x} = u_x = \sin(2yt) + 6xt,$$

$$\blacktriangleright \frac{\partial u}{\partial t} = u_t = 2xy \cos(2yt) + 3x^2,$$

$$\blacktriangleright \frac{\partial^2 u}{\partial x \partial y} = u_{xy} = \frac{\partial}{\partial x} 2xt \cos(2yt) = 2t \cos(2yt).$$

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 $u_{xx} = \frac{\partial^2 u}{\partial x^2}$ etc.

Partial Differential Equations

A partial differential equation (PDE) is an equation involving *partial derivatives*. A solution to a partial differential equation is a function (of more than one variable) that satisfies the equation. Some well known PDEs include:

- ▶ $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ one dimensional wave equation.
- ▶ $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$ one dimensional heat equation.
- ▶ $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ two dimensional Laplace equation.
- ▶ $\frac{\partial^2 u}{\partial t^2} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$ two dimensional wave equation.
- ▶ $\frac{\partial u}{\partial t} = c^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right)$ three dimensional heat equation.
- ▶ etc...

Partial Differential Equations

As mentioned earlier, a solution to a partial differential equation is a function that satisfies the PDE, that is, the function makes the PDE true when substituted into it.

Example: Show that all the following functions are solutions to the two dimensional Laplace equation $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$:

- ▶ $u = x^2 - y^2$,
- ▶ $u = e^x \cos(y)$,
- ▶ $u = \ln(x^2 + y^2)$.

Partial Differential Equations

- ▶ Given $u = x^2 - y^2$, then $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 2 - 2 = 0$
- ▶ Given $u = e^x \cos(y)$, then $\frac{\partial^2 u}{\partial x^2} = e^x \cos(y)$ and $\frac{\partial^2 u}{\partial y^2} = -e^x \cos(y)$, hence $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$
- ▶ Given $u = \ln(x^2 + y^2)$, then $\frac{\partial u}{\partial x} = \frac{2x}{x^2 + y^2}$ so

$$\frac{\partial^2 u}{\partial x^2} = \frac{2(x^2 + y^2) - (2x)(2x)}{(x^2 + y^2)^2} = \frac{2y^2 - 2x^2}{(x^2 + y^2)^2}$$

By symmetry,

$$\frac{\partial^2 u}{\partial y^2} = \frac{2x^2 - 2y^2}{(x^2 + y^2)^2}$$

and hence $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

Partial Differential Equations

An ODE with initial conditions is called an initial value problem (IVP).

We will solve PDEs with both initial conditions *and* boundary conditions. Mathematically, initial conditions and boundary conditions are the same, however the distinction is usually made that initial conditions refer to imposing solutions at certain points in *time*, while boundary conditions refer to imposing solutions at the boundary of the *space* in which we are interested.

A PDE with boundary conditions and initial conditions is called a *boundary value problem* (BVP).

Partial Differential Equations

Example:

Consider the one dimensional wave equation

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}.$$

Suppose the region we are interested in solving this over is $[0, 1]$

Suppose we impose an initial condition $u(x, 0) = \sin(\pi x)$, and boundary conditions $u(0, t) = 0$ and $u(1, t) = 0$.

Show that $u(x, t) = (\cos(c\pi t) + \sin(c\pi t)) \sin(\pi x)$ is a solution to this boundary value problem.

Sketch the solution on paper then code a surface plot in MATLAB as shown on the next slide.

Use $A \cos(\theta - \phi) = A \cos(\theta) \cos(\phi) + A \sin(\theta) \sin(\phi)$ to show $\cos(c\pi t) + \sin(c\pi t) = \sqrt{2} \cos(c\pi t - \frac{\pi}{4})$.

The code

We create the surface plot as follows.

```
figure(1)
clf
x=linspace(0,1);
t=linspace(0,2);
[X,T]=meshgrid(x,t);
c=1; %for example
U=(cos(c*pi*T)+sin(c*pi*T)).*sin(pi*X);
surf(X,T,U);
xlabel('distance  $x$ ','interpreter','latex');
ylabel('time  $t$ ','interpreter','latex');
zlabel('height of wave function  $u(x,t)$ ','interpreter','latex');
title('Wavefunction in time and space','interpreter','latex');
```

Label axes and give titles. Using the $\text{\LaTeX}$ interpreter for labels is optional, but useful if you write in $\text{\LaTeX}$.

Surface Plot: Solution to PDE

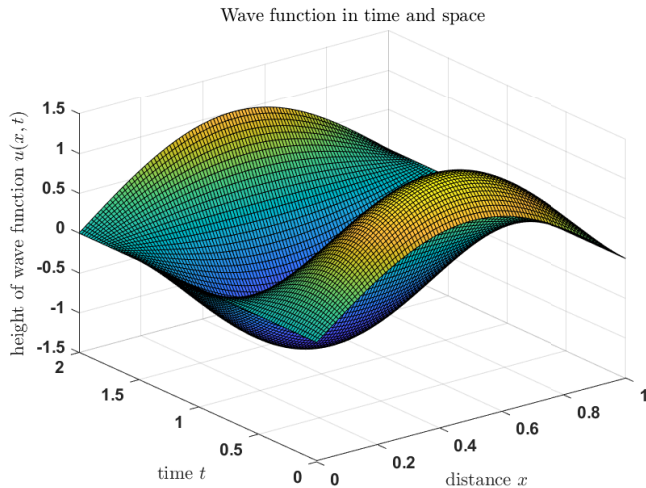


Figure: $u(x, t) = (\cos(c\pi t) + \sin(c\pi t)) \sin(\pi x)$, $c = 1$

Line Plot: Cross Section

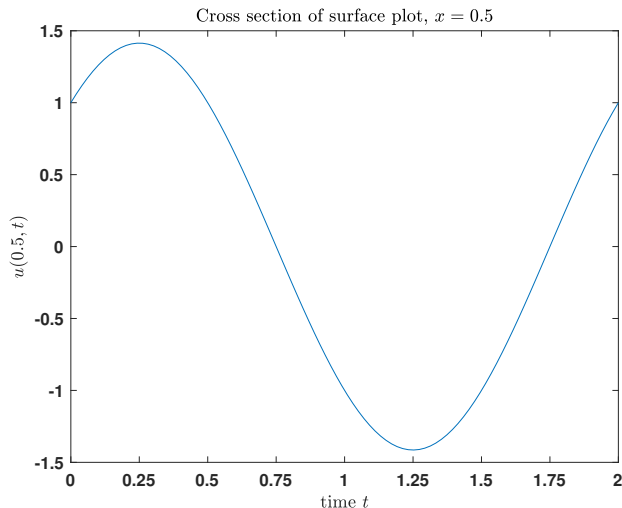


Figure: $u(0.5, t) = (\cos(c\pi t) + \sin(c\pi t)) \sin(\frac{\pi}{2}) = \sqrt{2} \cos(c\pi t - \frac{\pi}{4})$, $c = 1$

Partial Differential Equations

Now for two simple examples where we can solve PDEs like ODEs using techniques we already know:

Example:

Solve

$$\frac{\partial^2 u}{\partial x^2} - u = 0.$$

Example:

Solve

$$\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial x} = 0.$$

See textbook for more examples.

Partial Differential Equations

Solve for $u(x, y)$ given

$$\frac{\partial^2 u}{\partial x^2} - u = 0.$$

Since no y -derivatives occur, we can solve this PDE in the same way we would solve the ODE

$$\frac{d^2 u}{dx^2} - u = 0$$

to obtain

$$u(x) = Ae^x + Be^{-x}$$

with constants A and B . In our case, A and B may be functions of y giving

$$u(x, y) = A(y)e^x + B(y)e^{-x}.$$

Partial Differential Equations

Solve for $u(x, y)$ given

$$\frac{\partial^2 u}{\partial x \partial y} + \frac{\partial u}{\partial x} = 0.$$

As long as all the second derivatives of $u(x, y)$ are continuous then

$$\frac{\partial^2 u}{\partial x \partial y} = \frac{\partial^2 u}{\partial y \partial x}$$

Setting $\frac{\partial u}{\partial x} = p(x, y)$ we have $\frac{\partial p}{\partial y} = -p(x, y)$, $p(x, y) = c(x)e^{-y}$, and by integration with respect to x ,

$$u(x, y) = f(x)e^{-y} + g(y)$$

where

$$f(x) = \int c(x) \, dx$$

and $c(x)$ and $g(y)$ are arbitrary.

Exercises

Verify by substitution that the given functions are solutions of the one-dimensional wave equation, with suitable c . For each solution, sketch on paper and code as a surface plot.

$$u(x, t) = x^2 + t^2$$

$$u(x, t) = \cos(4t) \sin(2x)$$

Solve these PDEs like ODEs using techniques we already know:

$$\frac{\partial u}{\partial y} + y^2 u(x, y) = 0.$$

$$\frac{\partial^2 u}{\partial x^2} + 25\pi^2 u(x, y) = 0.$$

Text Book References

Chapter 12.1